

Real-Time Fraud Detection

Streaming features + sub-ms online serving

August 22, 2023

Headline Result

PR-AUC 0.974 (188x baseline); serving e2e p99
1.19 ms

Prepared for: Sai Veda

Publishing account: Nikeshk834

Project Context

Streaming feature engineering +
sub-millisecond online serving for card
fraud - imbalanced, latency-critical,
leakage-free.

This PDF dossier is intentionally visual and interview-ready: it captures the business problem, the technical architecture, the validation evidence, and the committed repository artifacts in a format that reads like a delivered project brief.

Repository slug

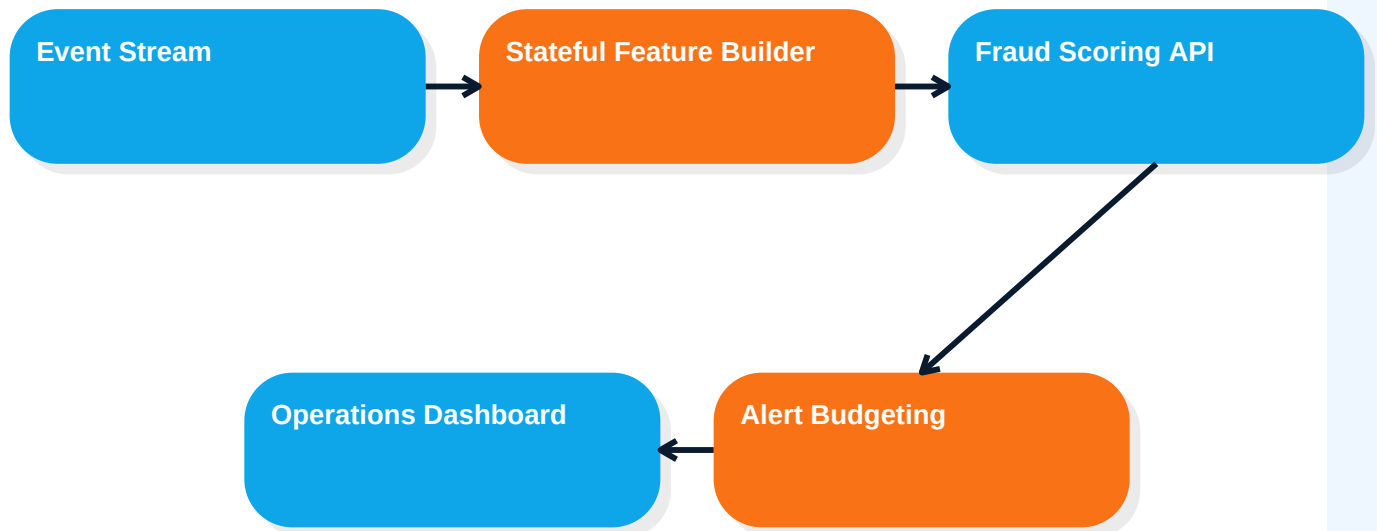
05-realtime-fraud-detection

Validation index

15 tests

Architecture Storyboard

The design optimizes for low-latency scoring with rolling behavioral features and a review budget that maps to real operations.



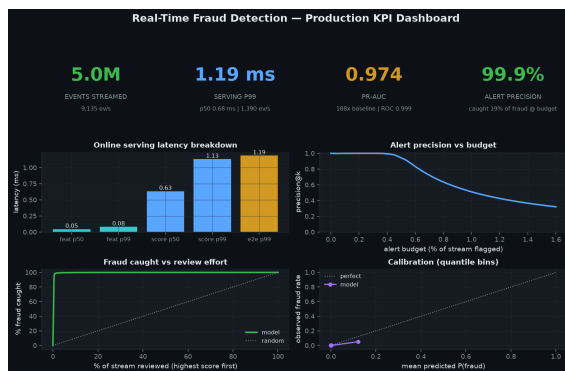
High-level focus area: The design optimizes for low-latency scoring with rolling behavioral features and a review budget that maps to real operations.

Validation And Visual Evidence

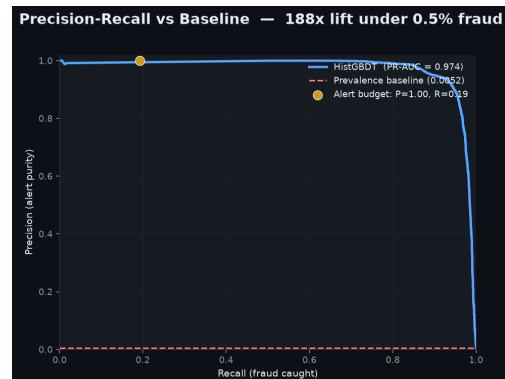
PR-AUC 0.974 (188x baseline); serving e2e p99 1.19 ms

15 passing tests

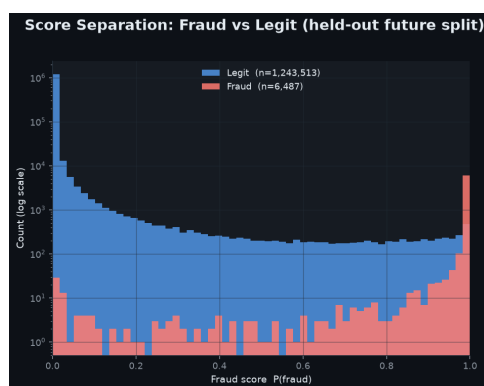
August 22, 2023



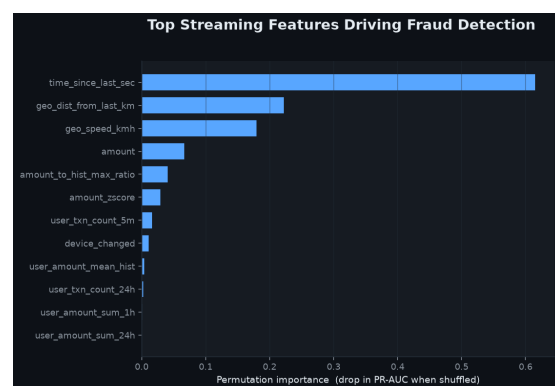
Production KPI dashboard



Precision-Recall vs baseline



Score distribution - fraud vs legit



Permutation feature importance

Repository Handoff

Prepared for: Sal Veda

Publishing account: Nikeshk834

Project date: August 22, 2023

Repository: 05-realtime-fraud-detection

Streaming feature engineering + sub-millisecond
online serving for card fraud - imbalanced,
latency-critical, leakage-free.

Committed deliverables

- README.md - product overview, quickstart, and measured headline numbers
- ARCHITECTURE.md - detailed design and scale trade-offs
- tests/ - automated validation suite
- benchmarks/ - measured run artifacts and result tables
- assets/ - generated dashboards, charts, and screenshots
- PROJECT_DOCUMENT.pdf - styled project brief for sharing and review

This document is intentionally more visual than the repository Markdown. It is meant to read like a polished project brief: business framing first, architecture second, proof third, and delivery scope last.